Sta 408

Mid term quiz for preparation;

(Searching file)

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Please If any mistake forgive me

1.Blocking refers to

(a) Hetrogeneous group of experimental units
(b) None of these
(c) Homogeneous group of experimental units
(d) Binding of homogeneous and heterogeneous experimental units in a group

2. if the total degrees of freedom and between treatments in a completely randomized design are 11 and 2 respectively,the degrees of the freedom within error will be.

(a) 11

(b) 2

(c) 3

(d) 9

3. A factor an experiment is

(a) experiment unit
(b) Controlled dependent Variable
(c) controlled independent variable
(d) treatments

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4. The Null hypothesis in a completely randomized design is

(a) Subtraction of all treatments means is equal to 1
(b) addition of all treatments means is equal to 1
(c) Not all treatments means are equal
(d) All treatments means are equal

5. Accepting SH_OS means that

(a) The sample means are different significancy

(b)The sample variance are different from each other

(c) The sample means are approximately equal

(d) The sample variance are approximately equal

6.The Randomizes complete block design controls the variability o=in

(a)No Variability

(b)One direction

(c) Two Direction

(d) Three direction

7.The efficiency of Randomized Complete Block Design (RCBD) relativs to Completely Randomized Design (CRD) can be obtained using the formula

(a)RE(RCBD,CRD)= $\frac{\text{Estimated MSE for RCBD-2}}{3\times\text{MSE for the CR design}}$

(b))RE(RCBD,CRD)= $\frac{\text{Estimated MSE for CR design}}{\text{MSE for the RCBD}}$ **(correct)**

(c) RE(RCBD,CRD)= $\frac{\text{Estimated MSE for (RCBD+CR design)}}{\text{MSE for the CR design}}$

(d) RE(RCBD,CRD)= $\frac{3\times\text{Estimated MSE for CR design}}{\text{MSE for the RCBD }+3}$

8.Analysis of variance is used to test the

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(a)Equality of three or more population means

(b) Equlity of two population Variance

(c) Equality of three or more population variance

(d)Equality of two population means

9. A factor is a ……..of treatments.

(a) category

(b) Design layout

(c) none of these

(d) Experimental error

10. Randomized complete block design is to adjust for the missing observation.

(a) Doubt full

(b) Easy

(c) Problematical

(d) Does not allow missing observations.

11.The variation among identically and independently treated experimental units is defined as

(a) Systematic error

(b) standard error

(c) experimental error

(d) Random error

12. If there are 7 treatments with 6 Randomized complete Block design then the degrees of freedom

For treatment

(a) 8
(b) 6
(c) 5
(d) 7

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13. To estimate the value of missing observation so that a complete set of data is obtained for carrying out the usual analysis of method is suggested by

(a)Gauss

(b) Yates

(c) Markav

(d) karl pearson

14. If there are k treatment with r blocks in a Randomized Complete Block Design for error will be

(a) k-1

(b) (n-k) (n-r)

(c) (k-1)(r-1)

(D) rk-1

15. A test is used in Randomized complete Block Design is

(a) F-test

(b) Z-test

(c) t-test

(d) chi square test

16. In a RCBD ,the variation is decomposed into

(a) Treatments and error

(b) Treatments and Blocks

(c) Treatments ,Blocks and Error

(d) Blocks and error

17. In systematic design, the treatments are applied to the experimental units by some

(a) Following other's wish

(b) None of these

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(c) Systematic pattern

(d) Random process

18. if the total degrees of freedom and between treatments in a completely randomized design are 11 and 2 respectively , the degrees of freedom within error will be

(a) 2

(b) 9

(c) 11

(d) 3

19. The term local of control refers to the

(a) Replication of experiment

(b) control of extraneous variations

(c) Amount of Balancing and grouping of experimental units

20.An experiment design is to be more efficient than another if it attains has same precision with

(a) Can never be efficient

(b) Can never be attain

(c) Greater expenditure of time and money

(d) Less expenditure of time and money

21.Formula by y= $\frac{rB+kT-G}{(k-1)(r-1)}$ **to find out the missing observation Randomized Complete Block Design, B shows**

(a) Sum of all values

(b) Number of replicates in one block

(c) Sum of remaining values in the blocks with the missing observation

(d) Sum of remaining values of the treatments with the missing observation

22. The F ratio in a completely randomized ANOVA is the ratio of

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(a) MSTR/MSE

(b) MSE/MSTR

(c) MSE/MST

(d) MST/MSE

23. One of the purpose of experimental design is to

(a) Increase the experimental error

(b) None of these

(c) Perform the statistical analysis

(d) Reduce the experimental error

24. The analysis of variance techniques are suitable to the designs only which are

(a) Systematic design

(b) Research design

(c) study design

(d) Randomized design

25.A doctor treats a patient with a skin condition with different creams to see which is most effective. In this example , a treatments is

(a) patient

(b) Different creams

(c) A doctor

(d)skin condition

26.A completely Randomized design is considered to be must useful in the situations when the experimental units are

(a) Hetrogeneous

(b) Homogeneous

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(c) Assorted

(d) None of these

27. The loss of information in CR design due to missing data compared to other design is

(a) None of these

(b) Equal

(c) Large

(d) small

28. One of the purpose of randomization is

(a) To remove the bias and other sources of extraneous variation

(b) To include as many as possible factors in the modal

(c)Reduce the cost

(d) Perform the statistical analysis

29. The randomization in a completely randomized design is restricted in

(a) Columns

(b) Rows

(c) Both rows and columns

(d) No restriction

30. In a basic ANOVA table, the Variation is decomposed into

(a) Between group

(b) within group

(c)Between and within group

(d) None of these

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31. The missing observations in the data impact on the error degrees of freedom as

(a) It is decreased by the number of missing observation computed
(b) Remains same
(c) It is two times increased by the number of missing observation computed
(d) It is increased by the number of missing observation computed

32. If in a CRD experimental units are heterogeneous then the design is

(a) Most sufficient

(b) Least efficient

(c)Most efficient

(d) Least sufficient

33. Experiment is performed on

(a) Sampling Frame

(b) Experimental unit

(c) Field unit

(d) Laboratory experimental

34. A factor is a ……..of treatments.

(a) Experimental Error

(b) category

(c) Design Layout

(d) None of these

35.A term of referring to the amount of balancing, blocking and grouping of the experimental units is

(a) Spurious effect

(b) Systematic error

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(c) Local control

(d) Extraneous Factor

36. If there are 7 treatments with 6 blocks in a Randomization Complete Block design then the degrees of freedom for treatments is

(a) 8

(b) 5

(c) 7

(d) 6

37. In systematic design, the treatments are applied to the experimental units by some

(a) Following other's wish

(b) Random process

(c) Systematic pattern

(d) None of these

38.In a RCBD the experimental error is decreased due to

(a) Experimental units itself

(b) The random allocation of treatments

(b) None of these

(d) Blocking of experimental units

39.The null hypothesis in a completely randomized design is

(a) Not all treatment means are equal

(b) All treatment means are equal

(c) Subtraction of all treatment means is equal to 1

(d) addition of all treatment means is equal to 1

40. A completely randomized design is applicable to a number of treatments

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(a)Specified

(b) All

(c) Small

(d) Large

41.Assume that each treatment "k" is replicated three times within each block, so that each block contains the plots

(a) n/k

(b) k/3

(c) 3k

(d) 9k

42. The random process of assigning treatments to the experimental units can be done by

(a) None of these

(b) Researcher's own choice

(c) Using random number table

(d) Repeating the basic experiment

43. In a Randomized complete block design, if a sum of square for blocks (SSB) is 9 with its corresponding degrees of freedom 3, the value of mean square block MSB will be

(a) 27

(b) 3

(C) 6

(d) 9

44. Experimental error may be due to the

(a) joint effect of treatments and experimental units

(b)Extraneous factors

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(c)All of these

(d)Natural variation among the experimental units

45.In a completely randomized design, a treatment sum of squares is SSTr and its corresponding degree of freedom is k-1,the error sum of squares is SSE with degree of freedom n-k, the mean square treatment (MSTr)will be

(a) SSTr/SSE

(b) SSTr/k-1

(c) k-1/SSTr

(d)SSTr/n-k

46.If the total degree of freedom in a completely randomized design is n-1 and the between treatments is k-1 then the degree of freedom within error will be

(a) n-k+2

(b)n-k-1

(c)k-2

(d) n-k

47. If there are 5 treatments and 5 blocks contain one missing observation in RCBD, then the degree of freedom for blocks is

(a) 4

(b) 5

(c) 3

(d) 6

48.In a randomized complete block design, a sum of square for treatment is SSTr, sum of squares for error is SSE and sum of squares total is SST.A formula to find sum of squares for blocks SSB by using other sum of squares is.

(a)SSB=SST-SSTr+SSE

(b)SSB=SST-SSTr-SSE

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(c)SSB=SST+SSTr-SSE

(d)SSB=SST+SSTr+SSE

49. In a completely randomized design, a mean square treatment MSTr is 6 and its corresponding degree of freedom is 3, the sum of square for treatments SSTr will be

(a) 2

(b)18

(c)9

(d)3

50.In a randomized complete block design with one missing observation, a sum of squares for error(SSE) is 12 and (k-1)(r-1)=4, the value of mean square error MSE will be

(a) 3

(b)4

(c) 6

(d) 8

51. In ANOVA the observation must be

(a) Independent

(b)Dependent

(c)Skewed to left

(d) Skewed to right

52. Planning and experiment to obtain appropriate data and drawing inference out of the data with respect to any problem under investigation is known as

(a) design and Statistical inference

(b)Probability and probability Distribution

(c)Laboratory Experiment

(d) Design and analysis of experiment

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53. To find the possible Latin squares of"k×k"Latin squares design, we use the formula

(a)k!(2k-1)!

(b)k!(k-1)!

(c)2k!(k-1)!

(d)k!(k-2)!

54. If there are 7 treatments with 4 blocks in a randomized complete block design (RCBD)

Then the degrees of freedom for error are

(a)19

(b)20

(c)25

(d)18

55.If there are 8 treatments and 4 blocks contain one missing observation in RCBD, then the error degrees of freedom is

(a)22

(B)20

(c)25

(d)36

56. The LSD test stand for

(a)Least squared Deviation test

(b)Least Sequential difference test

(c)Least significant Difference test

(d)Least squared Difference test

56. Efficiency of Latin square design can be compared with

(a) Completely Randomized Design (CRD)

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(b) Randomized Complete Block Design(RCBD)

(c)LSD_Test

(d) Both CRD and RCBD

57. For a 3×3 Latin square design,The standard Latin Squares can be

(a)one

(b) Two

(c)Three

(d) four

58. which condition must be hold for applying multiple comparison tests?

(a)Rejection of Alternative Hypothesis

(b) Rejection of null hypothesis

(c)Minimum mean Square Error
(d) Maximum Mean Square Error

59.The formula for the efficiency of Latin Sqaure design relative to RCBD when columns are removed as blocks,is

(a) $\frac{1}{k}\left[\frac{s_r^2}{s_e^2}+(K-1)\right]$

(b) $\frac{1}{k}\left[\frac{s_C^2}{s_e^2}+(K-1)\right]$ **(correct)**

(c) $\frac{1}{k}\left[\frac{s_C^2}{s_r^2}+(2K-1)\right]$

(d) $\frac{1}{k}\left[\frac{s_r^2}{s_e^2}+(K^2-1)\right]$

60.A teacher practice the different teaching methods on different groups in her class to see which yields the best results. In this example ,a treatments is

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(a)Different teaching methods

(b) Teacher

(c)Class

(d)Different groups

61.If the total degrees of freedom and between treatments in a completely randomized design are 15 and 4 respectively, the degrees of freedom for error will be

(a)11

(b)14

(c)18

(d)19

62.If there are 6 treatments with 3 blocks in a Randomized Completely Block Design (RCBD) then the degrees of freedom for error are

(a) 3

(b) 10

(c)6

(d)15

63.For a 6×6 Latin Square design there will be observation

(a)36

(b)6

(c)12

(d)24

64.Eight Fertilizer are being compared for their effects on a wheat variety. In this example treatments is

(a) Wheat

(b) Fertilize

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(c)Variety

(d) Effect

65.In a Latin Square design , number of rows and number of columns are

(a)Number of rows is greater

(b) Always Equal

(c)Number of columns is greater

(d) No specified rule

66. For a "4×4" Latin square design, the columns sums are 8,10,12,6 and rows sums are 11,6,12,7. The Grand total G will be.

(a) 27

(b) 36

(c) 72

(d)80

67.In a "K×k" Latin squares design, the degrees of freedom for error is

(a) (k-1)(k-2)
(b) (k-1)
(c) k(k-1)(k-2)
(d) (k-2)

68. if there are 4 rows 4 columns treatments formed a Latin Sqaure design, the total degrees of freedom will be

(a) 4

(b) 15

(c) 8

(d) 9

69.In a 5×5 Latin Square design, if the sum of squares for rows and columns is 16 and 12 respectively then the value of mean square columns is

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(a) 4.0

(b) 3.0

(c) 3.2

(d) 2.4

70.In a Randomized Complete Block Design, If a sum of squares for blocks (SSB) is 9 with corresponding degrees of freedom 3, The value of mean square block MSB will be

(a) 27

(b) 6

(C) 9

(d) 3

The End

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